

SHANTHA KUMAR

Full-Stack AI Developer - Bangalore, India

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PROFESSIONAL SUMMARY

Recent M.Sc. Information Technology graduate specializing in AI and Machine Learning. Skilled in Full Stack Web Development and Backend Engineering, with hands-on experience building AI-powered applications using LangChain, RAG, and FastAPI. Passionate about designing scalable APIs and responsive web applications to solve real-world challenges.

SKILLS

Frontend Development: HTML5, CSS3, JavaScript, React.js, Next.js, Tailwind CSS, UI/UX Fundamentals

Backend Development & Databases: Python, SQL, Django REST Framework (DRF), FastAPI, MySQL, PostgreSQL

Data Science & Machine Learning: NumPy, Pandas, Matplotlib, Data Analysis, Scikit-learn, Machine Learning Algorithms

AI & Generative AI: TensorFlow, RAG, LLMs, Gen AI, Prompt Engineering, Basics(CV, NLP)

Tools & Technologies: Git, GitHub, Docker, Postman, Microsoft Excel, Power BI, Visual Studio Code

PROFESSIONAL EXPERIENCE

Data Science Intern | Team PTID-AIE-JUN-26-11182

Coimbatore, TN | Jun 2026 – Present

- Identified and resolved structured data inconsistencies through rigorous Exploratory Data Analysis (EDA) and data preprocessing for projects, performing thorough root-cause analysis to address anomalous records.
- Built, trained, and fine-tuned predictive machine learning workflows leveraging XGBoost and Scikit-Learn across diverse classification and regression tasks, optimizing feature engineering strategies to enhance prediction metrics.
- Established comprehensive test cases and validation procedures for iterative modeling cycles, ensuring strict reproducibility and facilitating technical peer reviews.

PROJECTS

Local Retrieval-Augmented Generation (RAG) System | Python, FastAPI, Ollama, ChromaDB, LangChain

2026

- Engineered a contextually aware RAG framework using localized LLMs to extract precise insights from unstructured text, validating output accuracy through manual test-case comparison.
- Integrated LangChain and ChromaDB for vector search; designed and ran functional tests across query scenarios to confirm retrieval accuracy before delivering responses via Next.js and FastAPI.

Dynamic Web Application & API System | Python, Django REST, React, PostgreSQL

2024

- Designed a multi-tier web application with role-based authentication; wrote unit and integration test cases to validate access-control logic across user roles.
- Applied OOP principles to build extensible database schemas and REST APIs, and manually reproduced and resolved defects identified during testing.

EDUCATION

Master of Science in Information Technology

Government Arts College (Autonomous), Coimbatore | Bharathiar University

- Year of Graduation: 2026 | CGPA: 7.16

Bachelor of Science in Information Technology

Government Arts College (Autonomous), Coimbatore | Bharathiar University

- Year of Graduation: 2024 | CGPA: 7.48

Higher Secondary Certificate (PCM with Computer Science)

Sri Vijay Vidyalaya Matric Hr Sec School, Krishnagiri

- Year of Graduation: 2021 | Perc: 87.5%

CERTIFICATION & LANGUAGE

- Artificial Intelligence Engineer – Datamites, Coimbatore (In Progress – Expected Sep 2026)
- PowerBI Data Analytics – Microsoft Edunet (NM), Coimbatore (Certified: 2024)
- Cloud Computing Fundamentals – IBM Edunet Skillsbuild, Coimbatore (Certified: Jul 2025)
- LANGUAGE** -Tamil, English